

MOD 304 Course Outline

MOD 304 — Advanced CGE Modelling

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	55 (indicative band 50–60)
Shape	5 parts · 20 sessions · 8 laboratories · 4 assignments · 1 examination · 1 final project
Model and data	CDCGE - AICTEM Country-Disaggregated CGE (ENVISAGE 10.01 structure) · GTAP 11 (v11b), reference year 2017, values in USD million
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	MOD 303, and preferably MOD 301 and MOD 302. Comfort with constrained optimisation, duality, and the reading of a large parameter file. Learners should already be able to read a termination status before a variable without being reminded.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

MOD 303 built a standard CGE model and ran it. This course takes the specification apart. It asks what the standard nesting assumes and what the alternatives would assume instead; what a factor supply elasticity does that a closure cannot; why this model carries parameters for land, natural resources and human development that MOD 303 never mentioned; and how a modeller chooses between specifications when the benchmark cannot arbitrate. It ends with systematic sensitivity analysis and model comparison — the two practices that separate a defensible CGE study from a plausible one.

The subject of this course is the model rather than the country, so it runs no new scenarios and produces no new result about the economy. What it does instead is read the whole parameter file, the whole factor set and the whole tax architecture of a working model, and ask of each entry where the number came from. The land share is where that question is hardest, and the course does not repair it: school gap M-13 records what is wrong, why a repair would be somebody else's judgement, and what it forbids being said. All 51 registered findings are in the appendix.

The rule the course is built on

A parameter that looks like data is the most expensive kind. Elasticities arrive in the same file as shares and tax rates, print to the same number of decimals, and carry no mark to say that one was measured and the other assumed. Separating the two is the work; a sensitivity design is how you show which of them the result depended on.

Who it is for

Modellers who already run a CGE and now have to defend its specification: the nesting, the elasticities, the factor treatment, the closure and the sensitivity design. Reviewers who must judge whether a published model's structure can carry the claim made from it.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	What advanced means, and the parameter file	01–03	01	7.1
P2	Factors: the set, supply, segmentation, land	04–07	02–03	10.4
P3	Demand, competition, taxes, households and trade	08–12	04–05	12.2
P4	Closure, calibration diagnostics and the solver	13–15	—	5.6
P5	Scenario architecture, sensitivity, validation, comparison	16–20	06–08	13.8
	Assignment and assessment time			6.0
	Total	20	8	55.0

The 20 sessions and 8 laboratories, in teaching order, and the subject matter of each:

P1 What advanced means, and the parameter file — Sessions 01–03 · 7.1 h · Laboratory 01

- **01 What Advanced Means** — Where MOD 303 stopped, and why; The standard specification as an accumulation of conventions; Which conventions are innocuous and which are not; The

parameter file as the real specification; What this model carries that MOD 303 never showed; How this course is assessed: specification, not operation

- **02 Nesting Architecture** — The nest as a separability assumption; Leontief at the top: what it forbids; The value-added nest and the factor set; Where energy sits, and why it matters; Capital-energy complementarity against substitutability; How to report a nest so a reader can disagree with it
- **03 The Parameter File, Read in Full** — Every entry, classified by what it governs; Live parameters and inactive modules; Three parameters worth arguing about; The identification argument, stated properly; Calibrated, assumed, estimated; What a reviewer asks for first
- *Lab 01 Read the Whole Parameter File* — What does this model actually carry, and where did each entry come from?

P2 Factors: the set, supply, segmentation, land — Sessions 04–07 · 10.4 h · Laboratory 02–03

- **04 The Factor Set** — The factors this model carries, and their shares; Why the capital share is high; Mixed income and the split that produced it; Labour segmented by skill; Land and natural resources as specific factors; What the segmentation does not give
- **05 Factor Supply Elasticities** — Supply elasticity against supply closure; η_{land} , η_{nrs} and ω_{land} ; The corner cases each sits between; The price margin and the quantity margin; Which sectors each parameter bears on; How to report one, and why almost nobody does
- **06 Factor Market Segmentation** — The four specifications and their horizons; Sector-specific capital and rental dispersion; Reading someone else's tables for the closure; What this database can and cannot segment; Wage differentials as calibration residuals; Why an employment result needs a supply parameter
- **07 The Land Share** — The observation: seven decimal places, six sectors; Why it cannot be a measurement; The natural-resource column, and what it rules out; What school gap M-13 forbids, and what it does not; Why the school reports rather than repairs; The general diagnostic, and where else to apply it
- *Lab 02 The Factor Intensity Table* — What does the value-added nest look like sector by sector?
- *Lab 03 Factor Supply Elasticities* — What do the land and natural-resource supply elasticities do?

P3 Demand, competition, taxes, households and trade — Sessions 08–12 · 12.2 h · Laboratory 04–05

- **08 Demand Systems** — Cobb-Douglas, LES and CDE compared; Why Cobb-Douglas is unusable in a low-income economy; The Frisch parameter and its cross-country source; Subsistence and supernumerary expenditure; What identifies each system; The demand system as the weakest link in incidence work
- **09 Imperfect Competition** — What perfect competition imposes; Mark-up pricing and the conjectural variation; Increasing returns, firm entry and love of variety; What a SAM can identify about any of it; Why these results are so much more dispersed; When the added structure is worth the added assumption
- **10 The Tax Architecture** — Statutory against effective rates; The complete instrument set, and what this database carries; Effective rates by sector, read in full; Where an added instrument's incidence comes from; Revenue recycling as a modelling choice; Why fiscal work needs a fiscal SAM
- **11 Multiple Households** — The representative agent as an aggregation; The six things a disaggregated block requires; Splitting factor income, and why that is the hard part; What it would inherit: mixed income and M-13; Three routes to a distributional result; Why this school links to a survey instead

- **12 Trade Elaborations** — Nested Armington and bilateral sourcing; Why this build cannot price a preferential tariff; Trade and transport margins as a produced service; Tariff-rate quotas and non-tariff measures; The large-country case and terms of trade; What one aggregated partner forbids
- *Lab 04 Leontief and Ghosh Side by Side* — What do the two inverses say, and where do they disagree?
- *Lab 05 The Tax Architecture* — Which instruments exist, at what effective rate, on what base?

P4 Closure, calibration diagnostics and the solver — Sessions 13–15 · 5.6 h

- **13 Closure as a Research Question** — The five choices, restated as claims; Matching closure to horizon; Matching closure to the institutional setting; The closure decomposition, and its interaction term; Why every attribution in this school is argued; The closure section of a model note
- **14 Calibration Diagnostics** — The diagnostic battery, item by item; The homogeneity test, and why it is never reported; Walras's law as a running check; Share-set closure at full precision; Why replication is not validation; What to do when each diagnostic fails
- **15 Inside the Solver** — The complementarity problem, with bounds; Newton steps, the path search and the merit function; What each termination status is telling you; Scaling, and when it is the right remedy; Distinguishing a solver problem from a model problem; The composition trap in this school's own grid

P5 Scenario architecture, sensitivity, validation, comparison — Sessions 16–20 · 13.8 h · Laboratory 06–08

- **16 Scenario Architecture** — A shock, a stack and a sequence; Order dependence and the interaction term; Shapley-style attribution and what it costs; Why a comparative-static model has no baseline; What a policy package can be decomposed into; Reporting a stack honestly
- **17 Systematic Sensitivity Analysis** — One-at-a-time, grid, quadrature, Monte Carlo; Choosing the distributions, and defending them; Which parameters should be swept, and which are; The convergence obstacle: failures are not random; The failed-design-point rule; Reporting a range, and what it is a range over
- **18 Validation** — Why benchmark replication is not validation; Back-casting, and what it needs; Comparison against reduced-form estimates; Sign, ranking and magnitude tests; Homogeneity as a validation check; The honest sentence, and why it beats a chapter
- **19 Model Comparison and Reproducibility** — The four sources of disagreement; The comparison protocol; Harmonising data, parameters and closure; This school's own two models of one economy; The reproducibility package; The standard the final project is marked against
- **20 The Two Inverses** — The extracted input-output block, read; Establishing a basis from the data, not the documentation; Why the forward linkage is twice the backward; Key sectors and the size problem; One structural property, three symptoms, three courses; What the GTAP aggregation destroyed
- *Lab 06 A Systematic Sensitivity Design* — What range would a defensible sweep produce?
- *Lab 07 The Grid as an Experiment* — What does a full factorial convergence study establish?
- *Lab 08 A Model Comparison Note* — Why would two models of this economy disagree?

What a graduate can do

1. Set out a full production nest, justify each level, and say what a different ordering would change.
2. Distinguish a factor supply elasticity from a factor market closure, and say which questions each can answer.

3. Read this model's complete parameter file and classify every entry by what it governs and where it came from.
4. Specify a demand system appropriate to a low-income economy and state its identification requirements.
5. Set out the full tax architecture of a CGE model, including the instruments this database does not carry.
6. Explain what imperfect competition and increasing returns add, and what they cost in identification.
7. Design a closure as a research question rather than as a setting, and defend the choice.
8. Run calibration diagnostics beyond benchmark replication.
9. Explain how a mixed complementarity solver works well enough to diagnose its failures.
10. Design and execute a systematic sensitivity analysis and report a range rather than a point.
11. Validate a model against something other than its own benchmark, and say what validation is available when back-casting is not.
12. Write a model comparison note that a second analyst can act on.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 140 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35.04 %	8 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The parameter register; The factor block, at full precision; The sensitivity design; The specification note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete specification review of a CGE model — this one, or one the learner is responsible for — delivered as a ten-section specification note plus the diagnostic scripts that produced it

The per-laboratory weights are an equal split of the laboratory component rounded to two decimals, so the column sums to 100.04 rather than 100. The component weights are the binding ones.

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data and no household record. Every elasticity in the parameter file is a default from the ENVISAGE distribution rather than an estimate for this economy, and the course's position is that a default is a legitimate modelling choice and an illegitimate citation. The course performs neither the sensitivity sweep nor the alternative-closure run it teaches: it specifies both and records that neither was executed, which is the honest state of the artefact rather than a gap in the teaching.

Materials and set-up

A 125-page Technical Manual, 21 lecture decks of 212 slides, 8 laboratory decks of 99 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 140-item quiz bank, 5 assessment files, 19 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

CDCGE at D:/EDIP/models/CDCGE, the PATH licence, and Julia with `-pkgimages=no`. Laboratory 7 runs a grid and wants an hour of wall clock on a teaching machine; the Instructor Guide says how to cut it and what is lost by doing so.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in `COURSE_VERSION.json`.

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